Execution Oracles Set the Reported Risk of Conformal Abstention for Text-to-SQL [Experiment, Analysis & Benchmark]

Supplementary Material

Jiamiao Liu
Dewen Qiao
Jiajie Xu
Yu Zhang
Xuetao Chen*
Department of Information
Xinqiao Hospital, Army Medical University (Third Military Medical University)
Chongqing, China
xuetao@tmmu.edu.cn

This supplement accompanies the paper of the same title. Sections, tables and figures numbered without a letter are those of the paper.

A ENDPOINT EVOLUTION AND PREREGISTRATION HISTORY

Three preregistrations preceded the final preregistered panel. Table A1 lists what each froze and what it found. Apart from the analyses that the two expert audits preregistered (Sections G and H), every analysis after the preregistered result of Table 2 is post hoc. Each preregistration document carries the content hashes of the frozen code and data together with its deviation record. Each registration was written after the results of the previous stage were known and before it generated any new pool; the twelve main pools and four control pools of the panel were generated after the third registration and had not been sampled before it. The endpoint and the benchmark were therefore chosen during exploratory development on the same benchmark with one checkpoint, and the final panel is a fresh-generation confirmation of that endpoint on four checkpoints.

The saturated regime. On official Spider dev with XiYanSQL-QwenCoder-32B-2504, the score saturates: in a 500-question pilot with 20 samples per question, 465 questions placed every sample in one execution class and the score took 11 distinct values. The preregistered control on all 1,034 dev questions with 50 samples changed suite-oracle marginal risk by −0.64 points (D minus A) and coverage by +2.17 points, with 6.5 percent of questions having a shipped-database class that the suite splits; both changes are below the three-point criterion, and the control tested whether the sample budget produces the Spider-Realistic effect: raising the budget from 20 to 50 samples moved the dev-set contrast from 0.26 to 0.64 points while the Spider-Realistic contrast on the same checkpoint was unchanged.

Deviations from Registration 3. Two deviations are on record: the candidate table listed OmniSQL-32B before XiYanSQL-QwenCoder-32B-2504 while the text named the latter as the lineage representative, so a literal reading of the table order gives outcome O1, and both readings are reported with O2 as the conservative one; and the OmniSQL-32B pools were generated with tensor parallelism two where the frozen configuration said four, a serving choice.

B THE TIE-RULE AUDIT

The frozen analyser and the archived per-question files break a tie for the largest execution class differently (Section 3.7), and every recomputation from the files, including both censuses and all audited ladders, uses the file-order rule. A released script reconstructs the frozen analyser's own choice on each tied decision from the per-question files and applies it, so a recomputation can take either convention rather than being confined to the one the file order forces. Table A2 quantifies the consequence. When the frozen choice is reconstructed by re-running the union-find on the tied questions only, 53 of 12,192 top-class decisions are tied; 51 of them return a different SQL under the frozen rule and 27 change a correctness label. The frozen populations are therefore 522 answers for the disagreement census, one case the audit never saw and one labelled case outside the population, and 2,771 answers over 740 cases for the full census, 15 cases never seen and 17 labelled cases outside. Recomputing the GAP and D−A of top-class mass under the frozen rule, preregistered and in every column of both audited ladders, with the never-labelled answers bracketed as all semantic errors and as none, changes no cell by more than 0.0841 points; the two brackets differ by at most 0.0631 points; no sign changes and no cell crosses the three-point line, although 83 of the 9,600 split-level thresholds and 66 answer rates do move. The same recomputation of the other five scores of the six-score comparison, preregistered and under the four census conventions, changes no cell by more than 0.2664 points and no sign, with the two brackets at most 0.1914 points apart. Seven cells of the two likelihood scores under a census convention, each within 0.1 points of the three-point line under the file order, cross it under at least one bracket, so the counts of Section 6.2 become 33, 13 to 14, 9 to 10 and 1 in place of 32, 15, 10 and 1. The preregistered numbers of Table 2 are the frozen analyser's; the audited ladders of Tables A5 and A6 are file-order recomputations, which is why their preregistered columns differ from it. Both bounds are on the unrounded means; rounded to the two decimals the tables print, the top-class-mass values differ by up to 0.09 points.

*Corresponding author.

Table A1: The registration history of the study. Each registration froze its data, models, endpoint and decision rule before it generated new pools; Registration 1 also included one pool generated before it. The last column reports the result.

Stage	Data and models	Endpoint and rule	Result
Original proposal (before any registration)	Official Spider dev, 1,034 questions, XiYanSQL-QwenCoder-32B-2504	Multi-instance intervention lowers selective risk by at least three points	Reduction 2.45 to 2.78 points on the four Spider-Realistic pools of Registration 1 and at most 0.52 on Spider dev; criterion not met; endpoint replaced by the marginal-risk quantities GAP and D−A on Spider-Realistic
Registration 1 (2026-09-04)	Spider-Realistic, 508 questions, the same checkpoint, official comparator, three new generation seeds (four pools with the existing one)	GAP > 0 in every pool; D−A ≤ -3 points	GAP 3.19 to 3.52 points in all four pools; D−A crossed the bar in two of four pools
Registration 2 (2026-09-04)	Five general-purpose candidates under an entry rule; database-grouped splits; false-schema control	Entry rule, then the same endpoints on every entrant	No general candidate entered (outcome O5); on the XiYanSQL checkpoint grouped splits kept GAP and halved D−A; the control passed
Registration 3 (2026-09-05)	Six SQL-specialist candidates, disjoint 200-question entry pilot, seeds 101, 202 and 303, question and database splits co-primary, aggregation rule and outcome table (Section 3.5)	GAP rule and D−A rule of Section 3.5	Four entrants; outcome O2 (Table 2)
Post hoc	Same pools	Six-score extension, robustness checks, both censuses, both ladders, the D minus A decomposition and the alignment finding	Sections 4 to 6 and 7.2

Table A2: The tie-rule audit. The frozen analyser breaks a top-class tie by the largest integer union-find key; the per-question files sort by the stringified key, so every recomputation from them takes the first-listed class. Impact bounds are over the GAP and D minus A cells of top-class mass under the preregistered labels and in every column of both audit ladders, and of the other five scores of the six-score comparison under the preregistered labels and the four census conventions, after recomputing under the frozen rule and bracketing the never-labelled answers in both directions.

Top-class decisions examined	12,192
Tied decisions	53
Tied decisions whose representative differs under the frozen rule	51
Tied decisions whose correctness label differs	27
Disagreement census: archived / frozen answers	521 / 522
Full census: archived / frozen answers (cases)	2,779 / 2,771 (742 / 740)
Largest change of a GAP or D minus A, top-class mass / other five scores (points)	0.0841 / 0.2664
Largest bracket width from never-labelled answers, top-class mass / other five scores (points)	0.0631 / 0.1914
Sign changes, top-class mass / other five scores	0 / 0
Cells crossing the three-point line, top-class mass / other five scores	0 / 7

C CENSUS DETAILS

Taxonomy. Semantic error: the returned query answers the natural-language question wrongly. Underspecified question: the question does not fix the point on which the two queries differ, and both readings are faithful to it; typical cases are ties at an extremum where `ORDER BY ... LIMIT 1` and `WHERE x = (SELECT MIN(x) ...)` differ, disagreement over whether to deduplicate, and ambiguity over whether "name" means a first name or a full name. Suspected reference-query defect: the reference query is, in the labeller's judgement, the worse rendering of the question. Synthetic-instance defect: the two queries differ only on values a real database of the schema would not contain, such as one country name mapped to several codes. Comparator artefact: the two results carry the same information and the comparator still judges them unequal. For semantic errors the labeller recorded one of wrong filter, wrong join, wrong aggregate, wrong grouping, wrong set operation, wrong order or limit, and wrong projection.

Batching and adjudication. The disagreement census was labelled in six batches of forty consecutive cases in the first pass and re-batched with a stride of six, with the taxonomy listed in reverse order, in the second; the full census used a stride of thirteen. The second pass on the full census was additionally told that 472 of the 742 cases already disagree on the shipped database, so that a synthetic-instance-defect label there would amount to calling the benchmark's own database defective, a stronger claim to be made only with reason. A third, independent session of the same model saw both labels and both rationales for every disagreement and could rule for either side or against both. Each pass could flag a case as borderline; 165 of 240 disagreement-census cases and 332 of 742 full-census cases carry at least one flag (Table A3).

Distribution by checkpoint and schema. In the full census the semantic-error share of each checkpoint's audited answers is 30.1 percent for Kwai-32B, 41.5 for Kwai-14B, 29.4 for Omni-32B and 42.8 for XiYan-32B. Suspected reference-defect cases concentrate in world_1 (68 cases, 26 questions), car_1 (40, 12) and dog_kennels

Table A3: The two censuses and the agreement of the two independent labelling passes, computed from the case-level labels. Labels were assigned by the two-pass adjudicated AI audit. Binary agreement collapses the five labels to semantic error versus not. In the four agreement and κ rows, the full-census value is over all 742 cases and the parenthesised value is over the 502 cases labelled for the first time in that census, the other 240 being carried over from the disagreement census with their labels.

	Disagreement census	Full census
Population	weak-accepted, suite-rejected A-cell answers	any A- or D-cell answer the suite rejects
Answers	521	2,779
Distinct cases	240	742
Questions	99	203
Five-label agreement	85.0%	89.5% (91.6%)
Binary agreement	92.9%	95.6% (96.8%)
Cohen's κ, binary	0.801	0.905 (0.935)
Cohen's κ, five-label	0.787	0.851 (0.873)
Cases with a borderline flag in either pass	165 (68.8%)	332 (44.7%)
Adjudicated disagreements	36	78
Unresolved after adjudication	0	0

Table A4: Labels assigned to the two censuses, by distinct case and by answer occurrence. Labels are AI-assigned. The lower block gives the error type of the cases labelled semantic error.

	Disagreement census		Full census	
Label	cases (240)	answers (521)	cases (742)	answers (2,779)
semantic error	65 (27.1%)	136 (26.1%)	288 (38.8%)	972 (35.0%)
suspected reference-query defect	42 (17.5%)	107 (20.5%)	213 (28.7%)	887 (31.9%)
underspecified question	93 (38.8%)	195 (37.4%)	184 (24.8%)	723 (26.0%)
synthetic-instance defect	40 (16.7%)	83 (15.9%)	47 (6.3%)	161 (5.8%)
comparator artefact	0 (0.0%)	0 (0.0%)	10 (1.3%)	36 (1.3%)
Error type of semantic-error cases				
wrong filter	24		129	
wrong grouping	24		30	
wrong join	3		44	
wrong aggregate	4		31	
wrong set operation	4		26	
wrong order or limit	0		19	
wrong projection	6		9	

(31, 7), with 11 schemas represented; Figure A1a gives the full distribution. Of the 288 semantic-error cases, 129 are wrong filters, 44 wrong joins, 31 wrong aggregates, 30 wrong groupings, 26 wrong set operations, 19 wrong orderings or limits and 9 wrong projections.

D INSTRUMENT CHECKS

Every main pool has a parse rate of at least 0.9985, and one of the 304,800 generations in the twelve main pools was truncated (Table 1). Under the false schema the suite top-1 accuracy of Kwai-32B, Kwai-14B, Omni-32B and XiYan-32B is 0.0033, 0.0, 0.0079 and 0.0512; the two Kwai checkpoints mostly decline to write SQL (parse rates 0.156 and 0.325); Omni-32B still writes a query for 70.7 percent of samples and XiYan-32B for 99.98 percent; and the current-practice certificate answers 0.0, 9.8, 8.7 and 11.0 percent of questions with a suite-oracle marginal risk of 0.0, 0.098, 0.087 and 0.089, within the preregistered expectation of $\alpha + 0.02$. Two sensitivity analyses accompany the preregistered result: under the budget denominator the question-split GAP and D−A change by at most 0.14 points while some secondary operating points change substantially, because unusable samples become implicit abstention mass; and under a custom canonicalising comparator in place of the official one the direction is unchanged and the magnitudes slightly smaller, for example −10.1 to −9.7 points of contrast for Kwai-32B against −10.4. Refining the threshold grids of all six scores five-fold changes GAP and D−A by at most 0.13 points.

The gold-free construction reads the row-order flag per compared pair, which makes the equivalence relation non-transitive on at most 11 questions per pool; the union-find takes the transitive closure, so a gold-free class is a connected component rather than a set of pairwise equivalent candidates. The candidates-only construction reads one flag per question and is therefore transitive, and it differs from the gold-free construction in nothing else, so the gap between the two, at most 0.01 points on any checkpoint, is the net sensitivity to reading the flag per pair together with the closure that reading forces.

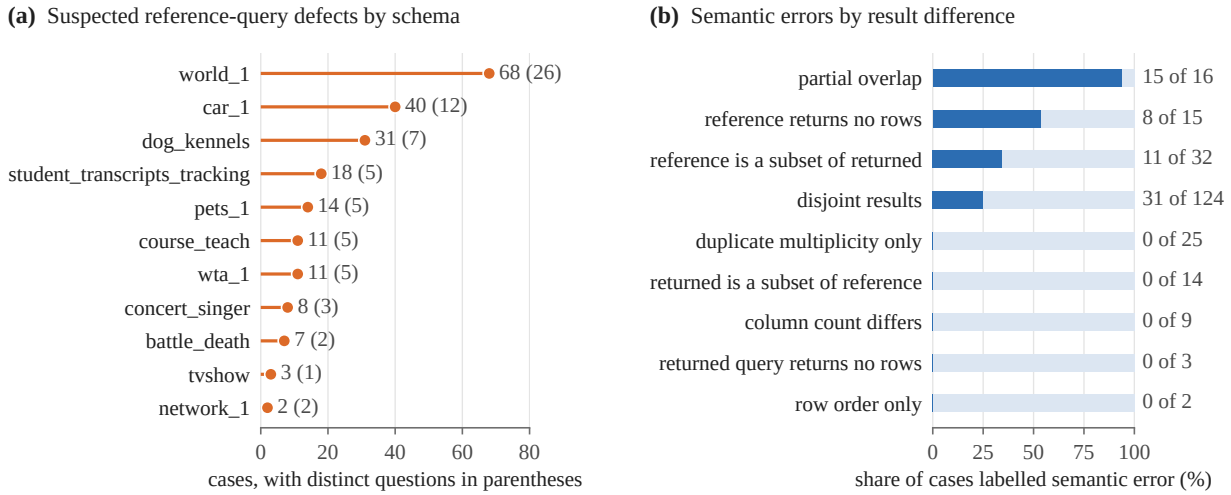


Figure A1: (a) Cases the AI audit labelled as a suspected reference-query defect in the full census, by schema, with the number of distinct questions in parentheses; the human audit of Table A11 gives a post-hoc design-weighted estimate of 17.5 percent of the 508 questions. (b) Share of cases labelled semantic error by the mechanical signature of the result difference on the first disagreeing instance, disagreement census; the signature is computed before any label is assigned.

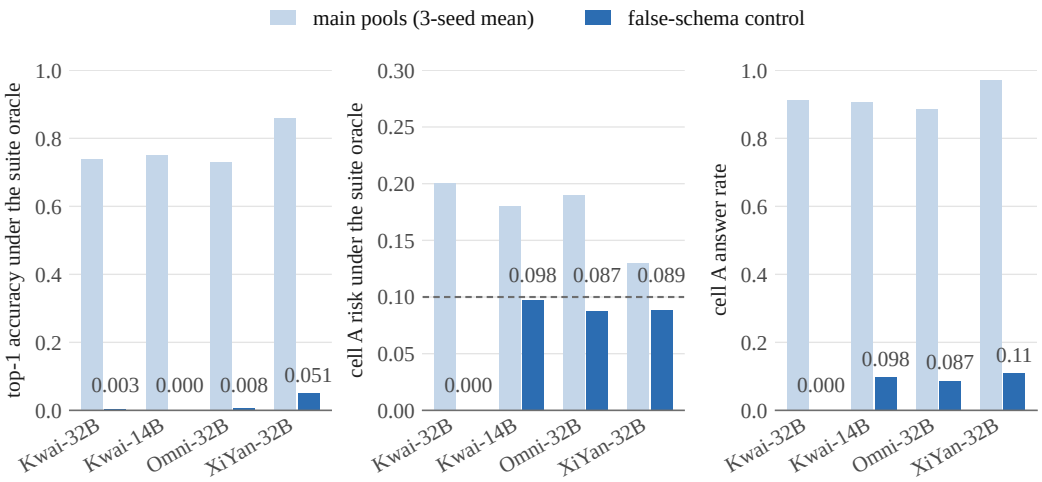


Figure A2: False-schema negative control. Each checkpoint was prompted with the schema of an unrelated database, one control pool per checkpoint with 20 samples per question and generation seed 999, against the three-seed means of the main pools with 50 samples per question; the control pools have 303, 481, 508 and 508 usable questions for Kwai-32B, Kwai-14B, Omni-32B and XiYan-32B, so the comparison is a diagnostic rather than a matched effect estimate. Top-1 accuracy under the suite oracle, the suite-oracle risk of cell A and its answer rate are shown. The certificate abstains on 89 to 100 percent of questions and keeps the suite-oracle risk at or below the nominal level plus 0.02, the preregistered expectation.

E ADDITIONAL FIGURES AND TABLES

This section collects the four cells of the intervention (Figure A3), the sweep over nominal levels (Figure A4), the empirical frontier (Figure A5), the six-score comparison (Figure A6), the exhaustive schema splits (Figure A7) and the appendix tables referenced from the main text.

F REPRODUCIBILITY

Code and data availability. The code and data of this study are released at https://github.com/yahiko-l/text-to-sql-oracle-certificates.

The code comprises the scripts that draw the candidate pools, execute and compare queries under either oracle, build the two partitions, calibrate and evaluate the certificate, and carry out every analysis reported here: the two pilot measurements on Spider dev,

Table A5: GAP (points) at $\alpha = 0.10$: the risk of the current-practice cell under the named yardstick minus the risk it reports on its own labels; only the preregistered column uses the unmodified suite oracle as the yardstick. Columns: under the preregistered suite labels, after relabelling the rejected side with the disagreement census (narrow convention), after relabelling it with the full census (narrow and unanimous conventions), and under the reverse stress test that additionally counts as wrong the 259 accepted answers that fall on questions with a suspected defective reference query. The stress test is an extreme assumption, not an estimate; the 1,580 distinct accepted outputs were never examined. Labels are AI-assigned. File-order recomputation (Section B).

Checkpoint	Split	preregistered	disagreement census	full census, narrow	full census, unanimous	reverse stress
Kwai-32B	question	+10.23	+2.26	-4.99	-7.51	-3.22
	database	+10.17	+2.21	-5.28	-7.81	-3.38
Kwai-14B	question	+8.08	+2.37	-3.37	-5.90	-0.24
	database	+7.87	+2.27	-3.28	-5.59	-0.26
Omni-32B	question	+9.50	+2.54	-4.25	-6.55	-2.18
	database	+9.49	+2.55	-5.17	-7.63	-2.91
XiYan-32B	question	+3.11	+1.00	-4.22	-7.15	+3.20
	database	+2.73	+0.81	-4.08	-6.51	+2.72

Table A6: The D minus A contrast (points, negative means cell D carries less risk under the named yardstick) at $\alpha = 0.10$ under the preregistered suite labels and under four nested relabelling conventions of the full census that exculpate rejected answers labelled as not semantic errors: wide (semantic error and synthetic-instance defect count as wrong), narrow (only semantic error counts as wrong), agreed (narrow, restricted to cases both passes agreed on), unanimous (agreed and neither pass flagged the case as borderline). The suite oracle stays the intervention; only the yardstick changes. The right block is the three-seed mean share of the 200 splits in which the contrast is negative, a resplit-stability statistic. Over the six scores of Section 4.3, the number of the 48 score-by-checkpoint-by-split cells that clear the preregistered three-point bar is 32, 15, 10 and 1 under the four conventions. Labels are AI-assigned. File-order recomputation (Section B).

		D−A (points)					share of splits negative				
Checkpoint	Split	prereg.	wide	narrow	agreed	unanim.	prereg.	wide	narrow	agreed	unanim.
Kwai-32B	question	-10.38	-3.92	-2.80	-2.05	-1.34	1.000	1.000	1.000	0.995	0.972
	database	-10.69	-4.10	-3.08	-2.32	-1.60	1.000	1.000	0.998	0.990	0.980
Kwai-14B	question	-8.30	-4.32	-3.74	-3.36	-2.54	1.000	1.000	1.000	1.000	0.998
	database	-8.02	-4.17	-3.65	-3.27	-2.61	0.997	0.997	0.993	0.993	0.992
Omni-32B	question	-9.58	-3.85	-2.71	-2.25	-1.35	1.000	1.000	1.000	0.998	0.963
	database	-10.21	-4.13	-3.14	-2.68	-1.68	1.000	0.997	0.988	0.982	0.963
XiYan-32B	question	-2.90	-1.66	-1.05	-0.99	-0.68	0.973	0.943	0.803	0.800	0.710
	database	-1.25	-0.73	-0.43	-0.46	-0.35	0.702	0.577	0.485	0.473	0.437

Table A7: GAP and D minus A (points) for six confidence scores under one certificate, one answer rule, the same pools and the same 200 splits, at $\alpha = 0.10$ under the preregistered suite labels. The top-class-mass row is the pipeline of the main text; the four consistency scores share the execution-equivalence relation under test; the two likelihood scores never read any oracle. All 48 GAP cells are positive and all 48 D minus A cells are negative. File-order recomputation (Section B).

	question splits				database splits			
Score	Kwai-32B	Kwai-14B	Omni-32B	XiYan-32B	Kwai-32B	Kwai-14B	Omni-32B	XiYan-32B
GAP (points)								
top-class mass	+10.2	+8.1	+9.5	+3.1	+10.2	+7.9	+9.5	+2.7
discrete semantic entropy	+9.8	+8.1	+8.7	+3.2	+9.7	+7.9	+9.0	+2.8
degree measure	+10.1	+8.1	+9.2	+3.1	+10.1	+7.9	+9.4	+2.8
set count	+8.8	+7.4	+7.7	+3.2	+8.7	+7.1	+8.1	+2.8
mean seq. log-prob	+8.4	+7.1	+6.7	+3.1	+8.3	+6.8	+7.4	+2.8
max seq. log-prob	+9.1	+6.6	+6.8	+3.0	+9.1	+6.4	+7.5	+2.7
D−A *(points)*								
top-class mass	-10.4	-8.3	-9.6	-2.9	-10.7	-8.0	-10.2	-1.2
discrete semantic entropy	-10.0	-8.2	-9.0	-2.9	-10.4	-8.0	-9.6	-1.2
degree measure	-10.2	-8.2	-9.3	-2.9	-10.5	-7.9	-10.0	-1.2
set count	-11.8	-11.5	-8.5	-3.7	-11.7	-10.3	-9.5	-2.0
mean seq. log-prob	-8.5	-7.3	-7.0	-2.9	-8.9	-7.1	-7.8	-1.6
max seq. log-prob	-9.3	-6.8	-7.1	-3.0	-9.9	-6.9	-7.8	-1.4

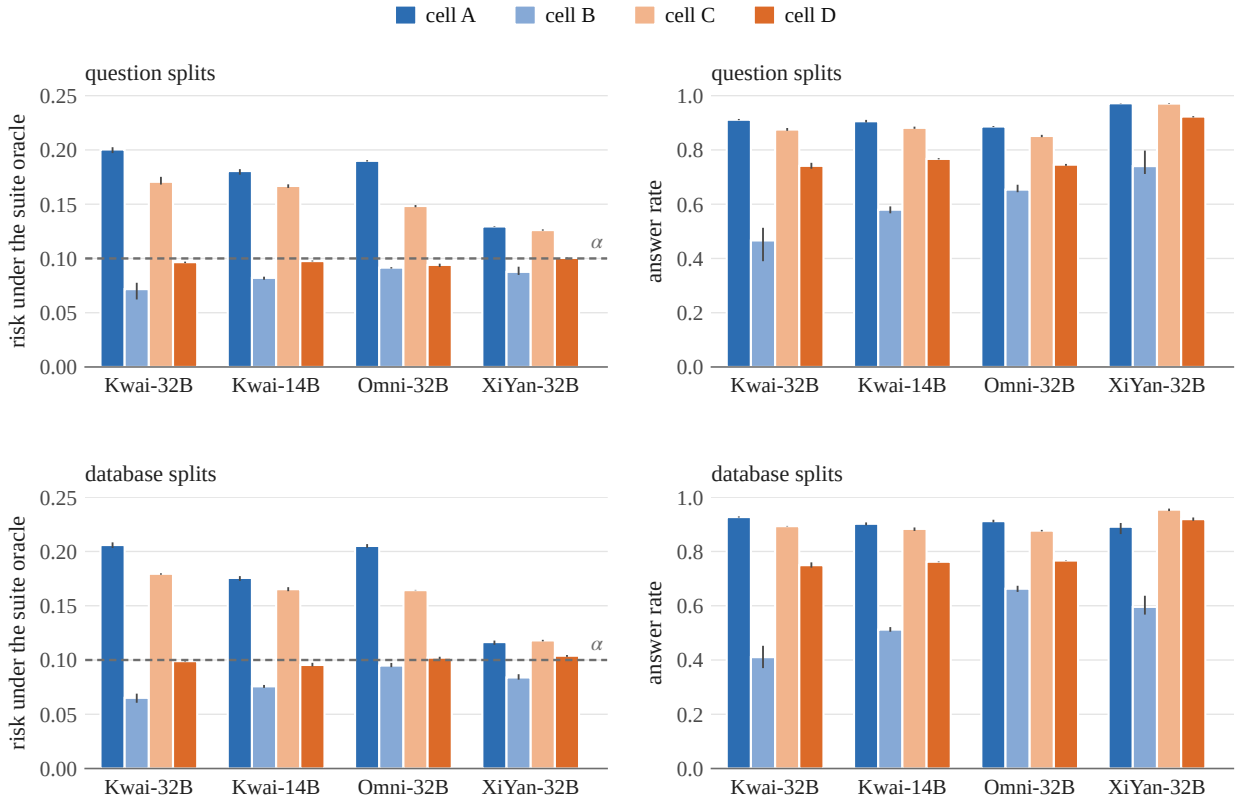


Figure A3: Risk under the multi-instance suite oracle and answer rate of the four cells of the intervention at nominal risk 0.10, three-seed means with the range over seeds, under question splits (top) and database splits (bottom). Cell A is current practice (original-database partition and labels) and cell D is fully multi-instance; cells B and C change only the labels or only the partition. The dashed line marks the nominal level.

the panel entry rule, the oracle intervention and its aggregation, the alpha curves, the exhaustive database splits, the cross-fit, the gold-free construction, the six-score comparison, both censuses, both audit ladders, the decomposition of the D−A contrast, the reconstruction of the frozen tie convention, the agreement between the two experts of each audit, and the preregistered analyses of both expert audits together with their bounding and matched-population recomputations. The data comprises the prompts and all candidate pools, about half a gigabyte of sampled SQL; the per-question sufficient statistics and the result files computed from them; the likelihood cache the two likelihood scores read; the two census label sets with the dossiers they were assigned from and their adjudications; and the two expert audits with the packages the experts received, their keys and the answer sheets. Every script names the inputs it reads, and every quantity reported here was produced by running these scripts on exactly those inputs. Running the table and figure scripts on the released result files returns all fourteen generated tables byte for byte and produces all nine generated figures, the registration-history table of Section A being written by hand and the schematic of Figure 2 drawn in the LaTeX source. The benchmark, its distilled test suites and the official evaluator are distributed by their authors. The prompts and call traces of the AI labelling passes were not retained. The blind-audit item sheet is byte-reproducible from its key, whose content hash was recorded before any human label existed.

Environment. Candidates were sampled with vLLM 0.22.1 in bfloat16 at a context of 8,192 tokens on NVIDIA H100 80GB cards, with tensor parallelism one for Kwai-14B and two for the three 32B checkpoints. Everything after sampling runs on CPU: a query is executed through the Python 3.12 `sqlite3` module against SQLite 3.45, and two results are compared by the official test-suite comparator of Zhong et al. [14] under the row-order convention of Section 3.1. The released analysis scripts need only NumPy, and the scripts that typeset the tables and draw the figures also need Matplotlib.

G THE PREREGISTERED BLINDED HUMAN AUDIT

The human audit tests the suspected reference-defect label. All 73 flagged questions were included, together with two stratified random controls of 40 questions each that measure different things: the 130 questions whose rejected answers the census examined without assigning a reference-defect label test whether the flag discriminates, and the 305 questions none of whose answers the suite

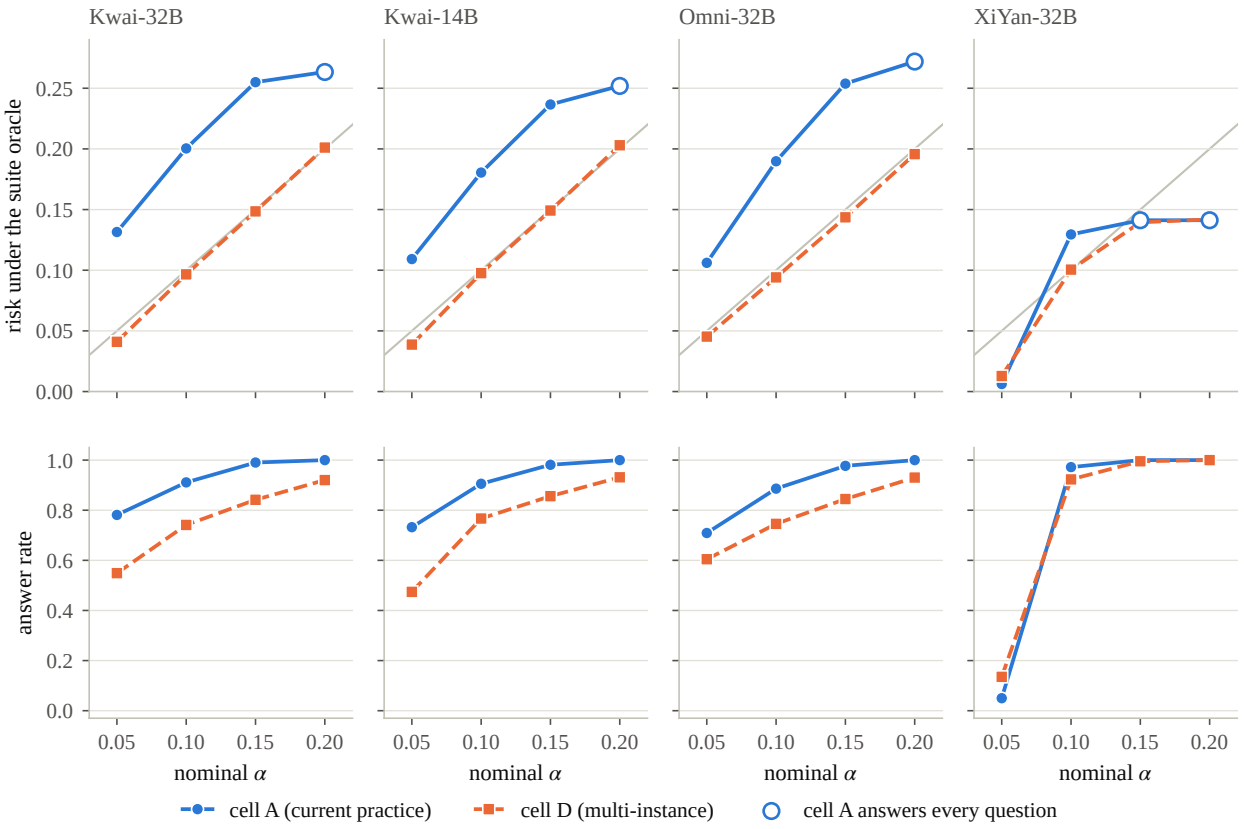


Figure A4: Suite-oracle risk (top) and answer rate (bottom) of the current-practice cell A and the multi-instance cell D against the nominal level, question splits, three-seed means. The grey line is the identity. Hollow markers show levels at which cell A answers every question, where the certificate has degenerated to answering everything and the ratio to the nominal level necessarily shrinks.

ever rejected measure the defect rate where these four checkpoints exposed nothing, which is the stratum a model-exposed count is silent about. Experts saw the question, the schema and the reference query only, in a shuffled order that carried no label vocabulary, stratum or question identifier, and were required to execute the reference query rather than read it, because the defect families of Section 5.4 are execution behaviour. There was deliberately no contrast pass showing model answers: the never-examined stratum has no suite-rejected answer by construction, so showing one where it exists would reveal the stratum. Two experts labelled independently and only their disagreements were adjudicated, by a third pass that saw both verdicts and both rationales and was free to return a verdict neither expert had given. The decision rule was fixed and committed before any human label existed: the flag is supported if the flagged stratum's defect rate exceeds the examined stratum's by at least 25 points and reaches 40 percent, refuted if the difference is under 10 points, and partially supported otherwise; on refutation the suspected-defect interpretation of this paper would have been withdrawn and only the oracle-relative sensitivity results retained. Secondary analyses, also preregistered, are the background rate with an exact binomial interval, agreement between human and AI labels, confirmation rates by the strength of the AI label, and the sensitivity of every primary quantity to including or excluding borderline verdicts.

Table A11 reports the outcome. The two experts agreed on 94.1 percent of the 153 items on the binary defective-or-not reading and on 91.5 percent across all four verdicts, with Cohen's κ of 0.863 on both readings. The third pass resolved their thirteen four-way disagreements, nine of which are also binary, ruling ten as the first expert had read them, two as the second had, and one as neither had. Under the adjudicated labels the flagged stratum was judged defective at 54.8 percent against 20.0 percent for the examined control, a margin of 34.8 points where the rule asks for 25, so the rule returns supported; dropping the verdicts marked borderline raises it further. That exclusion is convention-dependent, because twelve items carry the same verdict from both experts and different borderline marks, and the analysis reads the first expert's mark on them: it leaves 50 flagged and 25 examined items at a margin of 44.0 points, and excluding an item whenever either expert marked it leaves 42 and 22 at 60.2 points. Both clear the rule, and neither convention was preregistered.

In the never-examined stratum 3 of the 40 sampled questions carry a defective reference, 7.5 percent with an exact interval of 1.6 to 20.4. That stratum is defined by what the four checkpoints

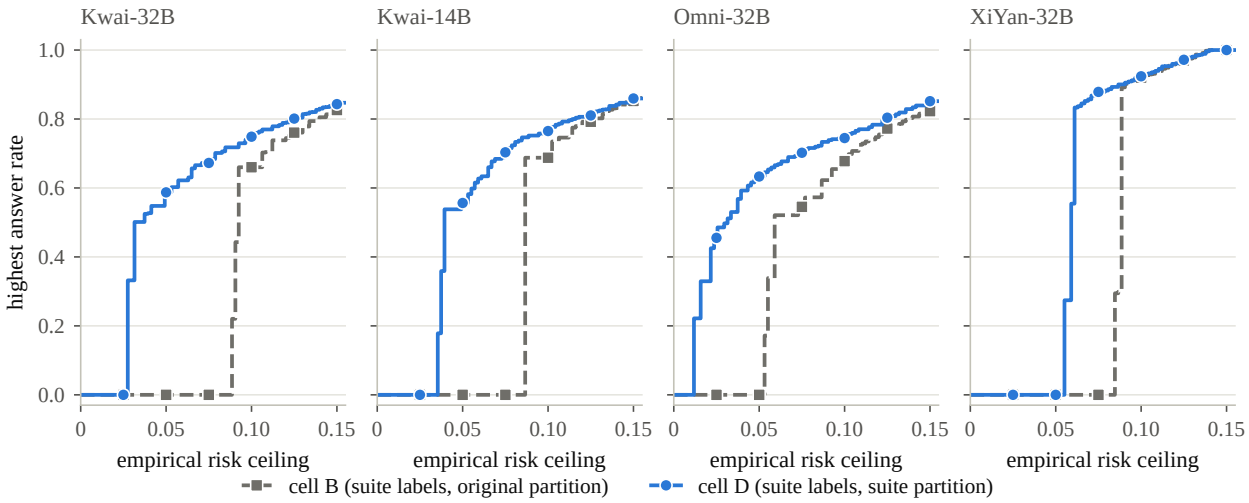


Figure A5: Highest answer rate reachable under a common empirical risk ceiling by cell B (suite labels, original-database partition) and cell D (suite labels, suite partition), three-seed means over all thresholds and the abstain-on-everything point. This is an empirical frontier on the full data, the two cells satisfy the same ceiling rather than the same realised risk, and it is not a held-out guarantee. The lines trace the frontier at every ceiling, and the markers give its value at the ceilings 0.025 to 0.15 in steps of 0.025.

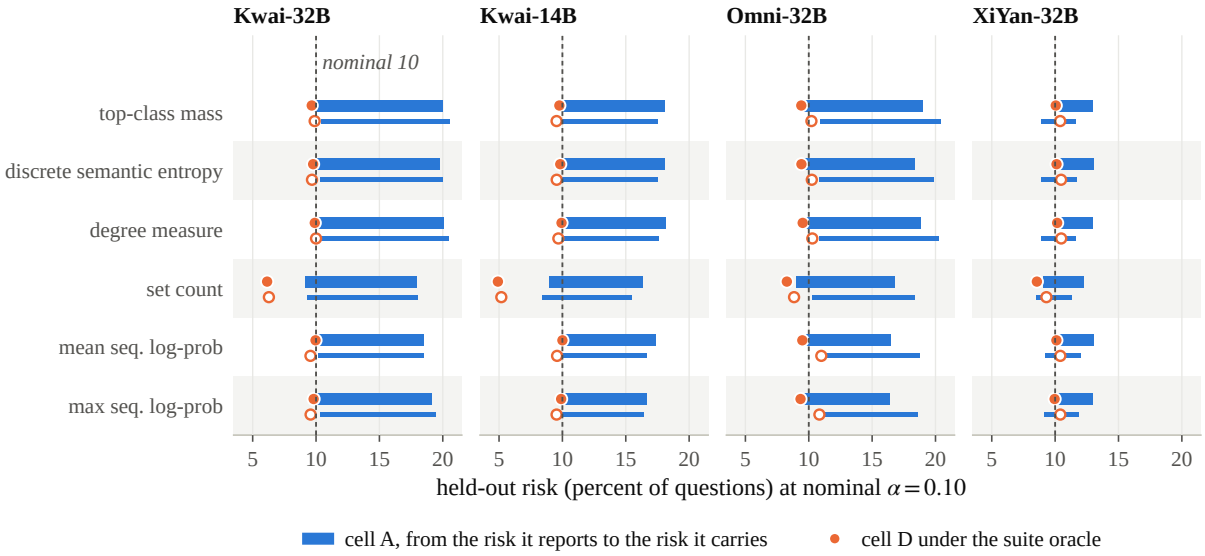


Figure A6: What the certificate reports and what it carries, at nominal risk 0.10 under the preregistered suite labels, for six confidence scores, four checkpoints and two split schemes, with one certificate, one answer rule, the same pools and the same splits. Each bar starts at the risk the current-practice cell A reports on its own labels, within 1.7 points of the nominal 10 in every cell, and ends at the risk it carries under the suite oracle, so the length of the bar is GAP; the circle is where the fully multi-instance cell D lands under the same oracle, and its distance back from the end of the bar is D minus A. The thick upper mark of each row carries the question-split value and the thin lower mark the database-split value. All 48 bars run to the right and every circle falls short of the end of its bar, whichever score sets the threshold.

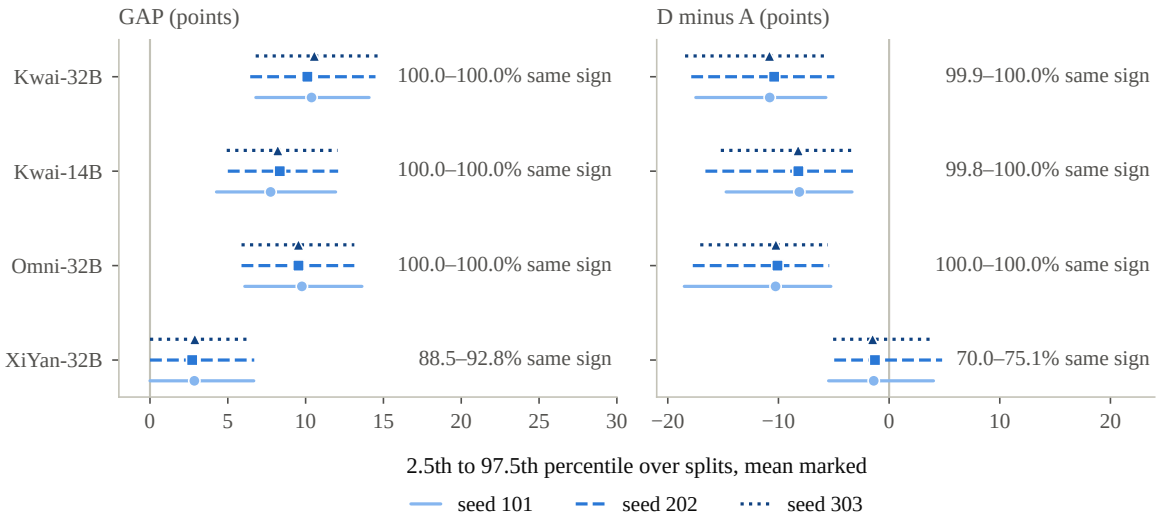


Figure A7: GAP and D minus A over all 92,378 schema-disjoint 9/10 splits of the 19 schemas, per checkpoint and generation seed: mean and the 2.5th to 97.5th percentile range over splits, with the range over seeds of the share of splits carrying the reported sign.

Table A8: AUROC of each score for predicting the correctness of the returned answer, computed on all 508 questions of each seed and averaged over the three seeds; the value does not depend on the split scheme. The score is built once on the original-database partition (cell A) and once on the suite partition (cell D), and each is evaluated under weak labels (correctness on the shipped database) and under suite labels. The drop is the weak-minus-suite change in AUROC in points for each cell; the same-SQL drop restricts the computation to the questions on which both cells return the same SQL, 477 to 508 per seed. The two likelihood scores use the same oracle-independent formula in both cells.

		Built on A		Built on D		Drop		Same-SQL drop	
Checkpoint	Score	weak	suite	weak	suite	A	D	A	D
Kwai-32B	top-class mass	0.846	0.743	0.804	0.839	+10.4	-3.5	+9.8	-3.1
	discrete semantic entropy	0.842	0.743	0.804	0.837	+9.9	-3.3	+9.4	-3.0
	degree measure	0.845	0.743	0.805	0.840	+10.2	-3.4	+9.7	-3.1
	set count	0.815	0.733	0.777	0.807	+8.2	-3.0	+7.8	-2.9
	mean seq. log-prob	0.711	0.682	0.710	0.689	+2.8	+2.1	+2.0	+2.0
	max seq. log-prob	0.712	0.661	0.712	0.666	+5.1	+4.5	+4.4	+4.4
Kwai-14B	top-class mass	0.838	0.754	0.810	0.828	+8.4	-1.8	+8.7	-1.0
	discrete semantic entropy	0.839	0.754	0.812	0.826	+8.4	-1.4	+8.8	-0.8
	degree measure	0.838	0.754	0.813	0.829	+8.4	-1.6	+8.8	-0.9
	set count	0.819	0.744	0.787	0.799	+7.6	-1.1	+7.8	-1.1
	mean seq. log-prob	0.740	0.681	0.727	0.672	+5.9	+5.5	+6.0	+6.0
	max seq. log-prob	0.696	0.645	0.684	0.634	+5.1	+4.9	+5.0	+5.0
Omni-32B	top-class mass	0.834	0.782	0.780	0.850	+5.2	-7.0	+4.2	-6.9
	discrete semantic entropy	0.831	0.783	0.777	0.849	+4.7	-7.3	+3.7	-7.2
	degree measure	0.833	0.783	0.779	0.851	+5.1	-7.2	+4.0	-7.1
	set count	0.802	0.767	0.753	0.814	+3.5	-6.2	+2.4	-6.0
	mean seq. log-prob	0.651	0.657	0.644	0.655	-0.6	-1.0	-1.1	-1.1
	max seq. log-prob	0.667	0.677	0.657	0.672	-0.9	-1.5	-1.4	-1.4
XiYan-32B	top-class mass	0.668	0.653	0.706	0.744	+1.5	-3.8	+1.7	-3.5
	discrete semantic entropy	0.670	0.655	0.710	0.747	+1.6	-3.7	+1.8	-3.4
	degree measure	0.669	0.653	0.707	0.745	+1.5	-3.8	+1.7	-3.5
	set count	0.672	0.656	0.711	0.747	+1.7	-3.6	+1.9	-3.4
	mean seq. log-prob	0.704	0.719	0.704	0.722	-1.6	-1.8	-1.5	-1.5
	max seq. log-prob	0.702	0.727	0.702	0.730	-2.5	-2.8	-2.4	-2.4

Table A9: Post-hoc robustness checks of the preregistered measurement (points, three-seed means, $\alpha = 0.10$, question splits unless stated). Instance cross-fit: classes and calibration labels from one half of each schema's suite instances plus the shipped database, correctness judged on the other half plus the shipped database, both directions (24 runs); the held-out D-cell risk row is its range. Candidates-only: partition built from the sampled candidates alone, the reference query used only to label a class afterwards. Gold-free: candidates-only with the row-order convention taken from the compared pair. Exhaustive database splits: mean over all 92,378 schema-disjoint 9/10 splits, with the range over seeds of the share of splits carrying the reported sign given in the two rows below. Leave-one-database-out: the preregistered question-split analysis with one schema removed. The matched original-Spider control (Kwai-14B, 508 one-to-one paired original questions) gives GAP +7.48 / +7.25 and D minus A -7.92 / -7.24 under question / database splits.

Check	Kwai-32B	Kwai-14B	Omni-32B	XiYan-32B
instance cross-fit GAP / D−A (24 runs)	+9.81 / -9.88	+7.84 / -7.96	+8.91 / -8.51	+3.11 / -2.90
instance cross-fit, held-out D risk (range)	0.094–0.101	0.097–0.099	0.096–0.101	0.100–0.101
candidates-only partition GAP / D−A	+10.23 / -10.38	+8.08 / -8.30	+9.51 / -9.59	+3.11 / -2.90
gold-free construction GAP / D−A	+10.23 / -10.38	+8.08 / -8.29	+9.51 / -9.59	+3.11 / -2.90
exhaustive database splits, mean GAP / D−A	+10.35 / -10.66	+8.10 / -8.18	+9.61 / -10.19	+2.81 / -1.39
exhaustive splits, share with GAP > 0 (range over seeds)	100.0–100.0%	100.0–100.0%	100.0–100.0%	88.5–92.8%
exhaustive splits, share with D−A < 0 (range over seeds)	99.9–100.0%	99.8–100.0%	100.0–100.0%	70.0–75.1%
leave-one-out GAP, min / max	+8.96 / +10.80	+6.97 / +8.95	+8.34 / +11.98	+1.17 / +3.86
leave-one-out GAP, schema whose removal gives the min	concert_singer	concert_singer	concert_singer	car_1
leave-one-out D−A, weakest (schema removed)	-8.98 (concert_singer)	-7.11 (concert_singer)	-8.34 (concert_singer)	-0.76 (car_1)

Table A10: Where the D minus A contrast comes from, at $\alpha = 0.10$, three-seed means over 200 splits. Both cells are judged by the same oracle, so a held-out question both cells answer with the same SQL contributes exactly zero and the three remaining events sum to the contrast; the identity is asserted on every split. The last column is the share of held-out questions on which both cells answer and the SQL is unchanged, which is not the 97.1 percent of Section 6.3: that share counts the representative of every question, answered or not. Post hoc. File-order recomputation (Section B).

			contribution (pts)			both answer,
Checkpoint	Split	D−A (pts)	changed SQL	only D	only A	same SQL (%)
Kwai-32B	question	-10.380	-0.001	+0.000	-10.379	74.1
	database	-10.695	-0.001	+0.023	-10.717	74.9
Kwai-14B	question	-8.297	+0.000	+0.001	-8.297	76.7
	database	-8.021	+0.000	+0.038	-8.059	76.1
Omni-32B	question	-9.575	-0.664	+0.033	-8.944	73.8
	database	-10.213	-0.854	+0.056	-9.415	75.6
XiYan-32B	question	-2.902	+0.005	+0.031	-2.938	91.8
	database	-1.246	+0.026	+0.338	-1.610	85.8

Table A11: Result of the preregistered blinded human audit of the reference queries. Experts saw the question, the schema and the reference query only, executed the query, and returned one of four verdicts per item. D is the share judged a defective reference query, with an exact binomial interval. The flagged stratum is a complete enumeration of the 73 flagged questions, so its interval describes uncertainty about a wider population of questions the flag could select rather than sampling error inside this set, while the two control intervals are sampling intervals for their own finite strata. The decision rule and its thresholds were committed before any human label existed.

Stratum	Items	Defective	D (%)	95% CI	Underspecified
flagged	73	40	54.8	[42.7, 66.5]	15
examined, not flagged	40	8	20.0	[9.1, 35.6]	9
never examined	40	3	7.5	[1.6, 20.4]	7
Preregistered verdict	supported, margin 34.8 points				
Excluding borderline	supported, margin 44.0 points on the first expert's marks, 60.2 if either expert's mark excludes				
Never-examined stratum	7.5% of its 40 sampled questions, 95% CI 1.6 to 20.4				
Human versus AI flag	agreement 71.2%, $\kappa = 0.416$, over all 153 items				
Between experts	binary agreement 94.1%, $\kappa = 0.863$; thirteen four-way verdict disagreements adjudicated, nine of them binary				

did not expose, so the figure is the defect rate among questions no answer of which the suite ever rejected, not a benchmark-wide prevalence: a defective reference is less likely to be caught where nothing was rejected. Weighting the three strata by their sizes gives the benchmark-wide point estimate, 17.5 percent over all 508 questions, which is post hoc and carries no preregistered interval. It is a stratified estimate in which only the two controls contribute sampling error, the flagged stratum being enumerated in full, and carries a design-based standard error of 2.7 points, an interval of 12.2 to 22.8 percent, three quarters of that variance coming from the never-examined stratum.

The human verdict and the AI flag agree on 71.2 percent of the 153 items, $\kappa = 0.416$. That figure counts the never-examined stratum as an AI negative, though the AI never looked at it; restricted to the 113 questions the AI did examine, agreement is 63.7 percent with $\kappa = 0.305$. Either way it is below what the two experts reach with each other, so the flag discriminates without being the same judgement. Confirmation tracks the strength of the AI label: 91.2 percent of the flagged questions whose label both AI passes assigned without marking it borderline were confirmed, against 27.6 percent of those the two passes merely agreed on and 10.0 percent of the ten that only the AI adjudication produced. Of the 51 references judged defective, the experts assigned a case-sensitive literal to 9, a wrong or negated filter to 7, numeric comparison or ordering on a text column to 7, a wrong grouping to 5, and two each to a wrong extremum, a missing join condition, an incomplete projection and a set operation keyed on a name; the remaining 15 fell outside the eight named families. Experts returned the underspecified-question verdict on 31 of the 153 items, 15 of them flagged, which is the preregistered secondary check on that reading of the questions; it is a judgement about a question alone, not about the point on which two queries differ, so it does not transfer to the census label of the same name.

Both experts are database engineers at tertiary hospitals with more than five years of SQL experience. They were recruited inside the laboratory and had not encountered this study before the package reached them: not the benchmark, not a model output, not a label of the AI audit, not the preregistration. They share a laboratory with the authors. The third pass over the thirteen disagreements was performed by the project lead. The preregistered category does not rest on it: run either expert's own sheet through the rule alone and it returns supported, at 36.2 points on the first and 27.7 on the second, with flagged rates of 56.2 and 45.2 percent against a threshold of 40. What the third pass decides is where inside that range the reported numbers fall. Its reach is bounded on both sides: no resolution of those thirteen items reaches refutation, and because the protocol lets the adjudicator return a verdict neither reviewer gave, the attainable margin runs from 20.2 to 38.7 points, so a reader who discounts the pass entirely still has partial support as the floor. Resolving the one item as neither reviewer had it lowered the margin rather than raising it.

Four limits bound what this establishes. The audit tested one label and tested it at the question level: the census attributes a defect to a particular reference query paired with a particular returned answer, and the experts saw no returned answer, so the 213 case-level attributions and the 887 answer occurrences they cover are supported only through the question-level flag. The semantic-error, underspecified-question, synthetic-instance and comparator-artefact labels of Section 5 remain AI-assigned. The experts were blind to the strata and to every model output but not to the hypothesis, and the protocol illustrated the defect families with families the AI audit itself had produced, each of which recurs among the flagged items. The protocol names no item and states that the families need not occur in the batch. And the protocol asks what a reference query returns on the supplied database, while six adjudications call a reference defective whose result on that database happens to be right and whose logic would fail on other data; scoring those six the other way gives 47.9 percent against 17.5 percent, a margin of 30.4 points, which the rule still supports.

H THE PREREGISTERED INDEPENDENT-YARDSTICK AUDIT

Section 7.2 leaves open whether the suite partition builds a better score or only looks better under the labels it produced, and neither oracle can settle that, so the audit brings in a third reading. Two SQL experts saw a question, its schema and one returned query, with no reference query, no score, no cell, no oracle label and no stratum, in an order shuffled under a fixed seed, and were required to execute the query rather than read it. The four verdicts are that the query answers the question, that it does not, that the question is underspecified so that both readings are faithful to it, and that the item cannot be judged; the last two are excluded from the endpoint and their share is reported. They cover 4.6 and 5.4 percent of the two sheets, and counting them as correct instead moves the contrast to 0.96 points on XiYan-32B and 7.08 on Kwai-32B, changing neither significance.

The endpoint is confined to the questions on which both cells return the same SQL, 507 of 508 on XiYan-32B and 496 of 508 on Kwai-32B, because there one verdict serves both cells and the cells differ only in the score. The 26 items covering the 13 questions where the returned SQL differs were judged as well and are reported separately: the experts agreed on 23 of them, 16 as answering the question and 5 as not. The sheet holds distinct answers rather than checkpoint-and-cell uses, and it deduplicates across checkpoints as well: on one question both checkpoints return the same SQL, so that answer was judged once and serves four uses, which is why the same-SQL items number 1,002 rather than 1,003 and the sheet totals 1,028. The endpoint then drops the items the experts disagree on and those whose verdict is not usable, so it rests on 463 of those questions on XiYan-32B and 455 on Kwai-32B (Table A12), and every suite-label contrast quoted against it is recomputed on those same questions: over all same-SQL questions the suite labels put the suite-partition score 9.2 points ahead on XiYan-32B and 8.9 on Kwai-32B, and on the questions the expert estimate retains those contrasts are 8.3 and 9.2, so the comparison in Section 7.3 changes the label without changing the population.

Two things were fixed before any label existed. The decision rule declares the outcome substantive when the suite-partition score has the higher AUROC on both checkpoints at $p < 0.05$, circular when neither does, and mixed otherwise. Those three names are the labels the registration gave to outcome patterns, and not

Table A12: The preregistered independent-yardstick audit. Two SQL experts, blind to the cell, the score and every oracle label, judged whether each returned query answers its question, on the 1,028 distinct answers of two checkpoints; they were required to execute rather than read. AUROC is for predicting a correct answer, for the score built on the original-database partition and for the same score built on the suite partition, against the same expert labels on the same questions. The contrast is paired and tested by DeLong. The experts agree on 971 of the items; the last row of each block is the widest range a hill-climbing search over assignments of the remaining 57 attained; significance does not change anywhere inside it. The decision rule was fixed before any label existed and returns MIXED.

Checkpoint	Labels	correct/wrong	AUROC on A	AUROC on D	Δ (pts)	p
XiYan-32B	both experts agree	370/93	0.560	0.575	+1.53	0.275
	first expert alone	378/105	0.561	0.583	+2.21	0.121
	second expert alone	385/96	0.555	0.567	+1.19	0.379
	searched assignments				+0.47 to +2.72	0.076 to 0.688
Kwai-32B	both experts agree	364/91	0.627	0.697	+6.96	0.004
	first expert alone	377/99	0.632	0.702	+7.08	0.002
	second expert alone	373/95	0.628	0.689	+6.09	0.009
	searched assignments				+4.18 to +8.83	0.000 to 0.045

Table A13: The certificate re-scored against the expert labels of Section H, at $\alpha = 0.10$ under question splits, generation seed 101. Unlike every earlier relabelling, these labels cover both sides of the suite oracle, so the sign is read off rather than bracketed. GAP is the risk the current-practice cell carries under the experts minus the risk it reports on its own labels; D minus A is the expert-judged risk of the fully multi-instance cell minus that of the current-practice cell, each cell answering at the threshold its own calibration selected. Rows differ in what is done with the questions the experts called underspecified or could not judge. Post hoc: the preregistration for this audit fixed the alignment endpoint and declined to preregister a claim about the sign.

Checkpoint	Unjudged verdicts	GAP (pts)	D−A (pts)
XiYan-32B	the suite label	+10.10	-2.42
	the shipped-database label	+9.25	-1.86
	dropped	+10.14	-1.42
	either expert alone	+9.41 to +11.93	-1.49 to -2.52
	experts call wrong 17.7% of the 407 answers the suite accepts, 36.8% of the 57 it rejects		
Kwai-32B	the suite label	+7.37	-4.57
	the shipped-database label	+6.19	-3.79
	dropped	+6.79	-3.58
	either expert alone	+6.42 to +7.94	-3.75 to -4.41
	experts call wrong 15.1% of the 352 answers the suite accepts, 36.3% of the 113 it rejects		

mechanisms this design identifies: a non-detection bounds the advantage on that checkpoint rather than establishing that the score carries none, so what the rule returns is read here as persistence or non-detection under an independent yardstick. And the two checkpoints were picked as opposite cases under the oracles' own labels, one symmetric and one not. The advantage persisted on the symmetric checkpoint, Kwai-32B, and was not detected on the other, XiYan-32B.

The audit differs from that of Section G in two respects: neither expert took part in that earlier round, and the protocol illustrates no defect family. The design calculation, run before the item sheet was frozen, put the paired standard error at 0.024 and the separable contrast at about 4.5 points; the realised standard errors are 0.014 and 0.024, so the audit resolves a contrast of that size, and on XiYan-32B the interval of Section 7.3, which runs to 4.3 points, excludes the 8.3 points the suite labels report. The 57 disagreements are bracketed by search rather than adjudicated, and the verdict does not change across the range the search attained.

Two preregistered secondary analyses of the mechanism. The first substitutes the expert labels for the suite's in the reversal test of Section 7.2, where the weak-minus-suite change in AUROC is positive for the score built on the shipped-database partition and negative for the score built on the suite partition, in 16 of 16 combinations. Replacing the suite labels by the expert labels removes the reversal on both audited checkpoints: the weak-minus-expert change is positive for both cells, 9.7 and 12.9 points on XiYan-32B and 22.0 and 12.9 on Kwai-32B, because the expert labels are harder for both scores and come from neither score's own oracle. The reversal is therefore a property of the relationship between the two oracles, not evidence that either partition builds the better score, which is the separate question Table A12 answers.

The second bounds the instance-level component of the alignment without any new label. The archived cross-fit pools build the suite partition and its labels from one half of each schema's suite instances and judge correctness on the other half, so a score can be read under the instances that built it and under instances it never saw. The suite-partition score loses more than the shipped-database score in six of the eight pools, the direction that alignment predicts, but the losses are 0.2 to 1.3 points of AUROC against an alignment effect near 9 points, and on XiYan-32B they are exactly zero because the two instance halves disagree on the label of one or two classes in about 1,255. Instance-level circularity, as this cross-fit measures it, is present and an order of magnitude smaller than the alignment.

What remains is carried by whatever the two halves share; the reference queries are the candidate explanation this design cannot separate from the benchmark's other shared assumptions, and an expert audit reaches that shared component where cross-fitting cannot.

I ADDITIONAL RELATED WORK

Tremante et al. [12] mined schema constraints so that the separating database a verifier returns is one a real deployment could hold, which is the concern that our synthetic-instance-defect label of Section 3.6 records case by case. Two of the works of Section 2 already examine what a stricter oracle rejects: besides the fifty counterexamples Klopfenstein et al. [7] inspected, Kim et al. [6] decomposed the false negatives of execution matching against expert judgement.

Conformal abstention for language models. Abbasi Yadkori et al. [1] calibrated an abstention rule on a similarity-based self-consistency score with a distribution-free bound on the hallucination rate, and calibrated against human ratings the threshold of the similarity match that labels their answers, so that the error the match reports approximately bounds the error the raters find; Quach et al. [11] calibrated the sampling and filtering of candidate generations so that the returned set contains an acceptable one with a guaranteed probability, and Mohri and Hashimoto [10] filtered the claims of a generation so that what remains is factual with a guaranteed probability. On the limits of such rules, Kotte [8] showed that, when the base risk μ of a generator exceeds α, any method that holds the risk of the answers it gives to α must abstain on a fraction of at least $(\mu - \alpha)/(1 - \alpha)$ of inputs, Xu et al. [13] calibrated abstention geometrically, and Marques F. [9] gave the law of the empirical coverage of split conformal prediction, which we use to read resplit variability.

Score implementations. Implementations of the score families of Section 4.3 are available in UQLM [2] and LM-Polygraph [4], and functional entropy transfers the semantic-cluster idea to code without executing it [3]. The five scores other than top-class mass are the black-box members of the text-to-SQL family of Section 2, tested here under one certificate.

J FURTHER DETAIL ON THE DESIGN

Selective risk. Besides the held-out marginal risk and the answer rate of Section 3.1, the analysis records the selective risk of each cell, the fraction wrong among answered questions, which selective classification in the sense of Geifman and El-Yaniv [5] controls and the certificate does not.

The near-miss census. For the second pilot of Section 3.2 we applied single-edit mutations to reference queries, dropping a conjunct, swapping an aggregate, flipping a comparison, dropping `DISTINCT`, changing the ordering direction and similar edits, executed the 856 mutants on every instance, and kept the 832 whose result differs from the original on at least one suite instance.

The panel. With the two candidates of Table 1 excluded, at 0.67 on the entry pilot and for producing no parseable output under two inference back-ends, the Meta lineage has no member in the panel.

The outcome table and the instrument. Besides the O1 and O2 of Section 3.5, the registration's outcome table names O4, if no new lineage has positive GAP, and O5, if no candidate enters. The instrument is checked at every tier by parse rate, truncation rate and the false-schema control, whose preregistered expectation is a suite top-1 accuracy at most 0.10 and a cell-A suite risk at most $\alpha + 0.02$.

The census dossier. For each case the mechanical step of Section 3.6 recorded the first instance on which the two queries disagree, the two result tables truncated to twelve rows, a difference signature, which is row order only, duplicate multiplicity only, different column count, one result a subset of the other, partial overlap, disjoint or one side empty, and surface features of both queries. The dossier presented each case with its question, both queries, the disagreement breadth and both tables, followed by the full schema, so that a labeller could judge without opening the benchmark.

K PER-SEED VALUES, NOMINAL LEVELS AND THE FRONTIER

Per-seed values. Generation seeds move the preregistered values of Table 2 little: over both split schemes, per-seed GAP spans 9.90 to 10.41 points for Kwai-32B, 7.51 to 8.30 for Kwai-14B, 9.44 to 9.63 for Omni-32B and 2.62 to 3.13 for XiYan-32B, and per-seed D−A spans −10.02 to −10.98, −7.88 to −8.47, −9.41 to −10.45 and −1.11 to −2.93 points.

Across the nominal level. At the levels of Figure A4, the suite-oracle risk of cell A is 1.8 to 2.0 times the nominal level at $\alpha = 0.10$ and 1.6 to 1.7 times at 0.15 for the three checkpoints other than XiYan-32B, and under both split schemes cell D sits at 0.039 to 0.051, 0.094 to 0.102 and 0.144 to 0.153 at those three lower levels. The current-practice cell of XiYan-32B saturates in both directions under question splits: at $\alpha = 0.05$ it answers 5.0 percent of questions and carries 0.006 suite-oracle risk, and from $\alpha = 0.15$ it answers everything.

The frontier. At matched nominal level cell B is the more conservative of the two cells calibrated on suite labels, answering 46.6 to 74.0 percent of questions against 74.1 to 92.3 percent for cell D at $\alpha = 0.10$ under question splits, while the two realised risks differ, 0.072 to 0.092 against 0.094 to 0.101, which is why the comparison of Section 4.2 is made at a common risk ceiling. On the frontier of Figure A5, at a ceiling of 0.075 three checkpoints still cannot answer under the label-only cell while the suite-partition cell answers 67.3, 70.3 and 87.9 percent, Omni-32B being the exception whose label-only cell answers 54.5 percent; at a ceiling of 0.10 the difference closes to 0.4 to 8.9 points, and at 0.125 and 0.15 it closes to 0.0 to 4.1 points.

L THE SIX SCORES IN DETAIL

Every score of Section 4.3 is a confidence score, with larger values meaning greater confidence, and the certificate answers when the score is at least the calibrated threshold. With $p_C = |C|/u_q$ the mass of class C among the u_q usable candidates of q: top-class mass is $\max_C p_C$; discrete semantic entropy enters as $\sum_C p_C \log p_C$, the negative entropy of the class masses; the degree measure is $\sum_C p_C^2$; the set-count measure enters as minus the number of classes; and

the two likelihood scores are the mean and the maximum over usable candidates of the candidate's cumulative token log-probability divided by its token count. Unusable candidates are removed before any score is computed. Each score has a fixed threshold grid that is never adapted to a split: 201 equally spaced points on $[0, 1]$ for top-class mass and the degree measure, 201 points on $[-\log 50, 0]$ for the entropy score, the integers -50 to -1 for the set count, and steps of 0.001 on $[-8, 0]$ together with $-\infty$ for the two likelihood scores.

Operating points. At $\alpha = 0.10$ the current-practice cell answers 72 to 98 percent of questions and carries 1.13 to 2.06 times the nominal level under the suite oracle across the 48 cells of Table A7, and the spread across scores is smaller than the spread across checkpoints.

Flip questions. On each split, a certificate's GAP is the number of held-out flip questions it answers as a share of all held-out questions, so a certificate that never abstains carries the held-out flip rate as its GAP. Under question splits the enrichment of flip questions among answered questions relative to their base rate is 84.0 to 102.5 percent, so for five score-and-checkpoint combinations the answered questions are slightly richer in flip questions than the pool. This does not decompose GAP into a label part and a score part, because GAP is a joint quantity of the labels and the answered set and different scores answer different questions.

M FURTHER ROBUSTNESS DETAIL

Instance cross-fit. In all 24 runs of the cross-fit of Section 4.4 the held-out suite risk of cell D lands at 0.094 to 0.101, so the contrast holds on suite instances the calibration did not see. The same check shows how redundant the suite is for this purpose: the two halves disagree on the label of at most 0.6 percent of single-instance classes, 1.9 percent of multi-instance classes and 1.6 percent of top classes in any run, which is why cross-fitting cannot stand in for semantic evidence.

Leaving a schema out. In the leave-one-database-out check of Table A9, the preregistered question-split analysis with one schema removed, leaving one schema out moves GAP by at most 2.5 points, on Omni-32B without world_1, and by at most 1.9 points for XiYan-32B, without car_1.

The matched control. The control of Section 4.4 that replaces every Spider-Realistic question by its one-to-one paired original Spider question keeps the schemas, the reference queries, the prompt template and the generation seeds.

The sample budget. GAP is a quantity of cell A alone, so drawing m of each question's usable candidates without replacement and recounting the shipped-database partition gives its score the resolution of an m-sample budget. Each class keeps the representative and the labels it has in the full pool, so a question's answer changes only when the thinning changes which class is largest, which happens for 1.3 percent of the questions drawn at $m < 50$; a full budget reduction would also re-select the most frequent query within each class. Over $m \in \{10, 20, 30, 50\}$ and both split schemes the three-seed mean GAP spans 0.45, 0.51, 0.81 and 0.51 points for Kwai-32B, Kwai-14B, Omni-32B and XiYan-32B, while the share of questions placing every candidate in one class falls by 2 to 15 points and the resolution of the score changes by a factor of three to five.

N WORKED EXAMPLES OF THE DEFECT FAMILIES

Beyond the two examples of Section 5.4, quoted from the audited cases with the benchmark's spacing before punctuation removed, the families are these. A join without a join condition: the reference for "What are the first name and last name of the professionals who have done treatment cheaper than average?" joins professionals to treatments without an `ON` clause, returning fifteen names where the joined query returns five. A negated filter: the reference for "What are the names of the dogs for which the owner spent more than 1000 for treatment?" selects the dogs `NOT IN` the set that spent more than 1000. An incomplete projection: "List the id and all lines" of an address is answered with two of the three address-line columns. A set operation on names: "students' first names who have both cats and dogs" is answered by intersecting first names, which merges different students who share one. And a reversed or missing restriction: "the weight of the youngest dog" is answered by the youngest pet of any type. The set operation on names and the missing pet-type restriction are among the families that return the same rows on the shipped database and are exposed only on suite instances that contain the collision.

Signatures. Of the signatures of Figure A1b, the case in which the reference's rows are a subset of the returned rows was labelled a semantic error 34.4 percent of the time. Five signatures never received that label: differing duplicate multiplicity only (25 cases), the returned rows being a subset of the reference's (14), a different column count (9), the returned query returning no rows (3) and row order only (2), together 53 cases and 95 answers that the suite rejects and the audit attributes to the question, the reference or a synthetic instance rather than to the system. Disagreement breadth points the same way: cases that disagree on at most 10 percent of the instances were labelled semantic errors 17.9 percent of the time, cases that disagree on more than half of the instances 31.7 percent.

O FURTHER DETAIL ON THE AUDITED LADDERS

Under the narrow convention the current-practice cell of Kwai-32B carries 0.0482 at $\alpha = 0.10$ under question splits, three-seed mean, against 0.0981 under its own weak labels. Answer rates are untouched by relabelling: the questions each cell answers are fixed by the calibration, and only their judged correctness changes. The D−A contrast is negative in 96.3 to 100 percent of splits for three checkpoints under every convention of Table A6, on average over seeds; for XiYan-32B under database splits the share of negative splits falls from 0.702 under the preregistered yardstick to 0.437 under the unanimous one.

The decomposition. In the decomposition of Table A10, changing which SQL is returned carries under 0.2 percent of the contrast for both Kwai checkpoints and for XiYan-32B under question splits, and for XiYan-32B under database splits it offsets 2.1 percent of the

contrast. Under the narrow convention and database splits, XiYan-32B answers 2.8 points more questions in cell D than in cell A and its contrast is −0.43 points.

P SCORE-LEVEL DETAIL OF THE ALIGNMENT COMPARISON

The AUROC levels behind Table A8 show that the labels also change the size of a lead. For XiYan-32B on the original-database partition the plain likelihood scores, 0.704 and 0.702 under weak labels and 0.719 and 0.727 under suite labels, beat every consistency score, 0.668 to 0.672 under weak labels and 0.653 to 0.656 under suite labels, and suite labels widen that lead from about three to about six points. The ordering is a property of the partition rather than of the scores: on the suite partition the same consistency scores reach 0.744 to 0.747 under suite labels, against 0.722 and 0.730 for the likelihood scores.

The two audited checkpoints. XiYan-32B was chosen because its oracle-internal pattern looked least like alignment, and most of the advantage the suite labels report for it is not visible under the independent labels of Section 7.3; Kwai-32B was chosen because its pattern looked most like alignment, and there an advantage persists under a yardstick that did not build it.

REFERENCES

[1] Yasin Abbasi Yadkori, Ilja Kuzborskij, David Stutz, András György, Adam Fisch, Arnaud Doucet, Iuliya Beloshapka, Wei-Hung Weng, Yao-Yuan Yang, Csaba Szepesvári, Ali Taylan Cemgil, and Nenad Tomasev. 2024. Mitigating LLM Hallucinations via Conformal Abstention. https://doi.org/10.48550/ARXIV.2405.01563

[2] Dylan Bouchard and Mohit Singh Chauhan. 2025. Uncertainty Quantification for Language Models: A Suite of Black-Box, White-Box, LLM Judge, and Ensemble Scorers. *Transactions on Machine Learning Research* (2025). https://openreview.net/forum?id=WOFspd4lq5 arXiv:2504.19254.

[3] Dylan Bouchard, Mohit Singh Chauhan, Zeya Ahmad, and Ho-Kyeong Ra. 2026. Functional Entropy: Predicting Functional Correctness in LLM-Generated Code with Uncertainty Quantification. https://doi.org/10.48550/ARXIV.2605.28500

[4] Ekaterina Fadeeva, Roman Vashurin, Akim Tsvigun, Artem Vazhentsev, Sergey Petrakov, Kirill Fedyanin, Daniil Vasilev, Elizaveta Goncharova, Alexander Panchenko, Maxim Panov, Timothy Baldwin, and Artem Shelmanov. 2023. LM-Polygraph: Uncertainty Estimation for Language Models. https://doi.org/10.48550/ARXIV.2311.07383

[5] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. https://doi.org/10.48550/ARXIV.1705.08500

[6] Heegyu Kim, Taeyang Jeon, Seunghwan Choi, Seungtaek Choi, and Hyunsouk Cho. 2024. FLEX: Expert-level False-Less EXecution Metric for Reliable Text-to-SQL Benchmark. https://doi.org/10.48550/ARXIV.2409.19014

[7] Rocky Klopfenstein, Yang He, Andrew Tremante, Yuepeng Wang, Nina Narodytska, and Haoze Wu. 2026. SpotIt: Evaluating Text-to-SQL Evaluation with Formal Verification. In *The Fourteenth International Conference on Learning Representations, ICLR 2026*. OpenReview.net. https://openreview.net/forum?id=iMkvR2ICSE arXiv:2510.26840.

[8] Varun Kotte. 2026. When Can Conformal Risk Control Certify LLM Outputs? Bounds, Impossibility, and Adaptation for Structured Generation. https://doi.org/10.48550/ARXIV.2606.29054

[9] Paulo C. Marques F. 2025. Universal distribution of the empirical coverage in split conformal prediction. *Statistics & Probability Letters* 219 (April 2025), 110350. https://doi.org/10.1016/j.spl.2024.110350

[10] Christopher Mohri and Tatsunori Hashimoto. 2024. Language Models with Conformal Factuality Guarantees. In *Forty-first International Conference on Machine Learning, ICML 2024*. OpenReview.net. https://openreview.net/forum?id=uYISs2tpwP arXiv:2402.10978.

[11] Victor Quach, Adam Fisch, Tal Schuster, Adam Yala, Jae Ho Sohn, Tommi S. Jaakkola, and Regina Barzilay. 2024. Conformal Language Modeling. In *The Twelfth International Conference on Learning Representations, ICLR 2024*. OpenReview.net. https://openreview.net/forum?id=pzUhfQ74c5 arXiv:2306.10193.

[12] Andrew Tremante, Yang He, Rocky Klopfenstein, Yuepeng Wang, Nina Narodytska, and Haoze Wu. 2026. SpotIt+: Verification-based Text-to-SQL Evaluation with Database Constraints. https://doi.org/10.48550/ARXIV.2603.04334

[13] Rui Xu, Yi Chen, Sihong Xie, and Hui Xiong. 2026. Geometry-Calibrated Conformal Abstention for Language Models. https://doi.org/10.48550/ARXIV.2604.27914

[14] Ruiqi Zhong, Tao Yu, and Dan Klein. 2020. Semantic Evaluation for Text-to-SQL with Distilled Test Suites. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 396–411. https://doi.org/10.18653/v1/2020.emnlp-main.29